Chapter 4

Cosets and Lagrange's Theorem

Road map

Let H be a subgroup of a group G. Multiplying every element of H by a fixed element of G produces a left or right *coset*. We shall prove carefully when two cosets are equal, show that cosets partition G, and count the pieces to obtain Lagrange's theorem. The chapter ends with complete left- and right-coset calculations in $D_3 \cong S_3$, including the example in which the two kinds of coset do not agree.

4.1 Left and right cosets

Definition 4.1: Cosets

Let G be a group, let $H \leq G$, and let $a \in G$. The *left coset* of H represented by a is

$$aH := \{ax : x \in H\},$$

whereas the *right coset* represented by a is

$$Ha := \{xa : x \in H\}.$$

The representative a need not belong to H. The notation simply means that we multiply every element of H by a, on the indicated side. In terms of the translations introduced in Chapter 2,

$$aH = \ell_a(H), \qquad Ha = r_a(H),$$

where $\ell_a(x) = ax$ and $r_a(x) = xa$.

If G is Abelian, then $ax = xa$ for every $x \in H$, and hence

$$aH = Ha.$$

For example, in the additive Abelian group $\mathbb{Z}$, if $H = m\mathbb{Z}$, then

$$k + m\mathbb{Z} = m\mathbb{Z} + k = \{k + mn : n \in \mathbb{Z}\}.$$

In a non-Abelian group, left and right cosets may be different. We will see an explicit example in D_3 at the end of this chapter.

Proposition 4.1: Every coset has the same size as the subgroup

For every $a \in G$, left translation restricts to a bijection

$$\ell_a|_H : H \longrightarrow aH, \qquad x \longmapsto ax.$$

Similarly, $r_a|_H : H \to Ha$ is a bijection. Consequently,

$$|aH| = |H| = |Ha|.$$

This statement is valid for finite and infinite sets; for finite sets it is an equality of ordinary integers.

证明. The map $x \mapsto ax$ lands in aH by the definition of aH, and it is onto because every element of aH has the form ax with $x \in H$. It is one-to-one: if $ax = ay$, the cancellation law gives $x = y$. Equivalently, its inverse is the restriction of $\ell_{a^{-1}}$, since

$$a^{-1}(ax) = x.$$

The proof for right translation is identical, with multiplication on the right. ∎

4.2 When are two cosets equal?

Although a coset is written using a representative, the representative is not unique. The following criterion is the basic computational tool.

Lemma 4.1: Equality criterion for cosets

For $a, b \in G$,

$$\boxed{aH = bH \iff a^{-1}b \in H \iff b^{-1}a \in H}$$

and

$$\boxed{Ha = Hb \iff ab^{-1} \in H \iff ba^{-1} \in H.}$$

证明. We first prove the criterion for left cosets.

Forward implication. Suppose $aH = bH$. Since $e \in H$, the element $b = be$ belongs to $bH = aH$. Thus $b = ax$ for some $x \in H$, and multiplication on the left by a^{-1} gives

$$a^{-1}b = x \in H.$$

This is the shortest proof. To reproduce the calculation in the handwritten notes, choose $x, y \in H$ such that $ax = by$. Then

$$a = byx^{-1}, \qquad b^{-1}a = yx^{-1} \in H.$$

Since a subgroup is closed under inverses,

$$a^{-1}b = (b^{-1}a)^{-1} \in H$$

as well.

Reverse implication. Suppose $a^{-1}b \in H$. Write $x = a^{-1}b$, so $x \in H$ and $b = ax$. If $y \in H$, then $xy \in H$, and therefore

$$by = a(xy) \in aH.$$

This proves $bH \subseteq aH$. Moreover,

$$b^{-1}a = (a^{-1}b)^{-1} \in H,$$

so the same argument with a and b interchanged gives $aH \subseteq bH$. Hence $aH = bH$.

The criterion for right cosets is obtained by the same argument on the other side. For example, $Ha = Hb$ implies $a \in Hb$, so $a = xb$ for some $x \in H$, and hence $ab^{-1} = x \in H$. Conversely, $ab^{-1} = x \in H$ means $a = xb$, from which $Ha \subseteq Hb$; using the inverse $ba^{-1} \in H$ gives the reverse inclusion. ■

A useful memory aid

For left cosets aH and bH, invert the element written on the *left*: test $a^{-1}b$. For right cosets Ha and Hb, invert the element written on the *right*: test ab^{-1}.

Example 4.1: Congruence classes in $\mathbb{Z}$

Take $H = m\mathbb{Z} \leq \mathbb{Z}$, using additive notation. The inverse of k is $-k$, so the left-coset criterion becomes

$$k + m\mathbb{Z} = \ell + m\mathbb{Z} \iff -k + \ell \in m\mathbb{Z} \iff m \mid (\ell - k).$$

Thus the cosets of $m\mathbb{Z}$ are precisely the familiar congruence classes modulo m.

4.3 Congruence modulo a subgroup

Definition 4.2: Left and right congruence

For a fixed subgroup $H \leq G$, define *left congruence modulo H* by

$$a \sim_L b \iff aH = bH \iff a^{-1}b \in H.$$

Define *right congruence modulo H* by

$$a \sim_R b \iff Ha = Hb \iff ab^{-1} \in H.$$

Proposition 4.2

Both $\sim_L$ and $\sim_R$ are equivalence relations on G.

证明. We give every step for $\sim_L$.

1. *Reflexivity.* For every $a \in G$,

$$a^{-1}a = e \in H,$$

so $a \sim_L a$.

2. *Symmetry.* If $a \sim_L b$, then $a^{-1}b \in H$. Since H contains inverses,

$$b^{-1}a = (a^{-1}b)^{-1} \in H.$$

Therefore $b \sim_L a$.

3. *Transitivity.* If $a \sim_L b$ and $b \sim_L c$, then $a^{-1}b \in H$ and $b^{-1}c \in H$. Closure of H gives

$$(a^{-1}b)(b^{-1}c) = a^{-1}(bb^{-1})c = a^{-1}c \in H.$$

Hence $a \sim_L c$.

For $\sim_R$, replace the tests above by $ab^{-1} \in H$. For instance, transitivity follows because

$$(ab^{-1})(bc^{-1}) = ac^{-1} \in H.$$

■

The equivalence class of a under $\sim_L$ is exactly aH:

$$[a]_{\sim_L} = \{b \in G : a^{-1}b \in H\} = \{ax : x \in H\} = aH.$$

Similarly, $[a]_{\sim_R} = Ha$.

Theorem 4.1: Cosets partition the group

The distinct left cosets of H are pairwise disjoint and their union is all of G. The same is true of the distinct right cosets.

证明. Every $g \in G$ lies in the left coset gH, because $g = ge$ and $e \in H$. Thus the union of all left cosets is G.

It remains to show that two left cosets cannot overlap without being equal. Suppose

$$x \in aH \cap bH.$$

Then $x = ah_1 = bh_2$ for some $h_1, h_2 \in H$. Rearranging,

$$a^{-1}b = h_1h_2^{-1} \in H.$$

The coset-equality criterion therefore gives $aH = bH$. Hence two left cosets are either equal or disjoint. The right-coset proof is analogous. ■

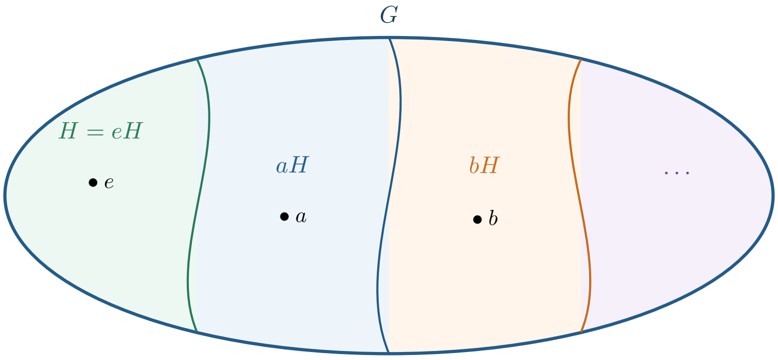

图 **4.1:** The subgroup H and its distinct left cosets form a partition of G. Each part has the same cardinality as H.

4.4 Coset spaces and index

Definition 4.3: Sets of cosets

The set of all left cosets of H in G is

$$G/H := \{aH : a \in G\}.$$

The set of all right cosets is

$$H\backslash G := \{Ha : a \in G\}.$$

A quotient set is not yet a quotient group

At this point, G/H means only a *set* of left cosets. A multiplication of cosets is well defined only under an additional condition—normality of H—which will be studied later. Also note the notation: the slash points away from the subgroup, so G/H contains left cosets and $H\backslash G$ contains right cosets.

Because $\mathbb{Z}$ is Abelian, left and right cosets of $m\mathbb{Z}$ coincide:

$$\mathbb{Z}/m\mathbb{Z} = m\mathbb{Z}\backslash\mathbb{Z}.$$

For example,

$$\mathbb{Z}/3\mathbb{Z} = \{\, 3\mathbb{Z},\ 1 + 3\mathbb{Z},\ 2 + 3\mathbb{Z} \,\}.$$

Definition 4.4: Index

The *index* of H in G, written

$$[G : H] \qquad \text{or equivalently} \qquad |G : H|,$$

is the number of distinct left cosets of H in G.

For a finite group, the number of left cosets equals the number of right cosets. This will follow immediately from the counting formula below. There is also a direct bijection

$$G/H \longrightarrow H\backslash G, \qquad aH \longmapsto Ha^{-1}.$$

Indeed, if $aH = bH$, then $a^{-1}b \in H$, and after taking inverses, $b^{-1}a \in H$. In particular $a^{-1}b \in H$, so the right-coset criterion applied to the representatives a^{-1} and b^{-1} gives $Ha^{-1} = Hb^{-1}$. The same formula applied twice gives the inverse map.

4.5 Lagrange's theorem

Theorem 4.2: Lagrange's theorem

Let G be a finite group and $H \leq G$. Then

$$\boxed{|G| = [G : H]\,|H|}$$

and therefore

$$\boxed{[G : H] = \frac{|G|}{|H|}.}$$

In particular, the order of every subgroup divides the order of the group.

证明. Let the distinct left cosets be

$$a_1H, a_2H, \dots, a_rH, \qquad r = [G : H].$$

By the partition theorem,

$$G = a_1H \sqcup a_2H \sqcup \cdots \sqcup a_rH,$$

where $\sqcup$ indicates a disjoint union. Every coset has $|H|$ elements because $h \mapsto a_ih$ is a bijection $H \to a_iH$. Counting the elements in the disjoint pieces gives

$$\begin{aligned} |G| &= |a_1H| + |a_2H| + \cdots + |a_rH| \\ &= \underbrace{|H| + |H| + \cdots + |H|}_{r \text{ terms}} \\ &= r|H| = [G : H]|H|. \end{aligned}$$

Dividing by $|H| \neq 0$ yields the formula for the index. ∎

Proposition 4.3: Orders of elements divide the group order

If G is finite and $a \in G$, then

$$\operatorname{ord}(a) \mid |G|.$$

证明. The cyclic subgroup generated by a is $\langle a \rangle \leq G$, and its order is precisely $\operatorname{ord}(a)$. Applying Lagrange's theorem to $\langle a \rangle$ gives

$$|G| = [G : \langle a \rangle]\,|\,\langle a \rangle\,| = [G : \langle a \rangle]\,\operatorname{ord}(a).$$

∎

Proposition 4.4: Groups of prime order

If $|G| = p$ is prime, then the only subgroups of G are $\{e\}$ and G. Moreover, every $a \neq e$ generates G, so

$$G \cong \mathbb{Z}_p.$$

证明. For any subgroup $H \leq G$, Lagrange's theorem says that $|H|$ divides p. The only positive divisors of a prime are 1 and p. If $|H| = 1$, then $H = \{e\}$; if $|H| = p = |G|$, then $H = G$.

Now choose $a \neq e$. The subgroup $\langle a \rangle$ contains both e and a, so it is not $\{e\}$. It must therefore be all of G. Thus G is cyclic of order p, and hence isomorphic to $\mathbb{Z}_p$. ■

4.6 Complete coset calculations in D_3

Recall the presentation

$$D_3 = \left\langle a, b \mid a^3 = e,\ b^2 = e,\ ba = a^2 b \right\rangle = \{e, a, a^2, b, ab, a^2 b\}.$$

Geometrically, a is a rotation through $120°$, while b is a reflection. The relation $ba = a^2 b = a^{-1} b$ expresses the fact that performing a rotation and a reflection in the opposite order reverses the rotation.

The complete Cayley table from the handwritten notes is

$$\begin{array}{c|cccccc}
\cdot & e & a & a^2 & b & ab & a^2b \\
\hline
e & e & a & a^2 & b & ab & a^2b \\
a & a & a^2 & e & ab & a^2b & b \\
a^2 & a^2 & e & a & a^2b & b & ab \\
b & b & a^2b & ab & e & a^2 & a \\
ab & ab & b & a^2b & a & e & a^2 \\
a^2b & a^2b & ab & b & a^2 & a & e
\end{array}.$$

4.6.1 The subgroup $H_1 = \langle a \rangle$

Let

$$H_1 = \{e, a, a^2\} = \langle a \rangle \cong \mathbb{Z}_3.$$

Lagrange's theorem gives

$$[D_3 : H_1] = \frac{|D_3|}{|H_1|} = \frac{6}{3} = 2.$$

Thus there must be exactly two left cosets. One is H_1; choosing $b \notin H_1$ gives the other:

$$\begin{aligned} bH_1 &= \{be, ba, ba^2\} \\ &= \{b, a^2 b, ab\}. \end{aligned}$$

Here $ba = a^2 b$, while

$$ba^2 = (ba)a = (a^2 b)a = a^2(ba) = a^2(a^2 b) = a^4 b = ab.$$

Therefore
$$D_3/H_1 = \{H_1, bH_1\}.$$

For the right cosets,
$$\begin{aligned} H_1 b &= \{eb, ab, a^2 b\} \\ &= \{b, ab, a^2 b\}. \end{aligned}$$

As unordered sets,
$$bH_1 = \{b, a^2 b, ab\} = \{b, ab, a^2 b\} = H_1 b.$$

Consequently,
$$D_3/H_1 = \{H_1, bH_1\} = \{H_1, H_1 b\} = H_1 \backslash D_3.$$

In this example every left coset of H_1 is also a right coset.

4.6.2 The subgroup $H_2 = \langle b \rangle$

Now take
$$H_2 = \{e, b\} = \langle b \rangle \cong \mathbb{Z}_2.$$

Its index is
$$[D_3 : H_2] = \frac{|D_3|}{|H_2|} = \frac{6}{2} = 3.$$

The three left cosets are
$$\begin{aligned} H_2 &= \{e, b\}, \\ aH_2 &= \{a, ab\}, \\ a^2 H_2 &= \{a^2, a^2 b\}. \end{aligned}$$

They are disjoint, each contains two elements, and together they contain all six elements of D_3. Thus
$$D_3/H_2 = \{H_2, aH_2, a^2 H_2\}.$$

The three right cosets are different:

$$\begin{aligned} H_2 &= \{e, b\}, \\ H_2 a &= \{a, ba\} = \{a, a^2 b\}, \\ H_2 a^2 &= \{a^2, ba^2\} = \{a^2, ab\}. \end{aligned}$$

The last equality uses the calculation $ba^2 = ab$ displayed above. Hence

$$H_2 \backslash D_3 = \{H_2, H_2 a, H_2 a^2\} \neq \{H_2, aH_2, a^2 H_2\} = D_3/H_2.$$

For example,
$$aH_2 = \{a, ab\} \quad \text{whereas} \quad H_2 a = \{a, a^2 b\}.$$

This is the promised concrete demonstration that left and right cosets need not coincide in a non-Abelian group.

What to retain from this chapter

1. Left and right cosets are translates: $aH = \ell_a(H)$ and $Ha = r_a(H)$. Translation is a bijection, so every coset has the same size as H.

2. The equality tests are $aH = bH \iff a^{-1}b \in H$ and $Ha = Hb \iff ab^{-1} \in H$.
3. Left and right congruence modulo H are equivalence relations. Their equivalence classes are the corresponding cosets, which partition G.
4. If G is finite, then $|G| = [G : H]|H|$. Hence subgroup orders and element orders divide $|G|$.
5. A group of prime order has no nontrivial proper subgroup and is cyclic.
6. In D_3, the cosets of $H_1 = \langle a \rangle$ agree on the left and right, whereas those of $H_2 = \langle b \rangle$ do not.